

6.2 Audio Encoders and Speech Models

Patch or frame the spectrogram, then a transformer. Whisper-style encoder–decoder.

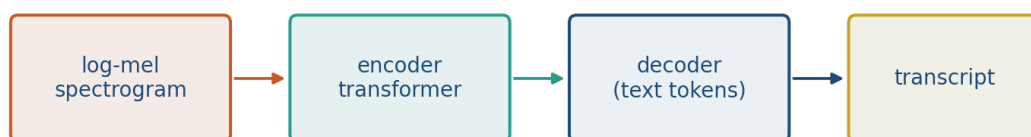
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Two audio systems you will cite: **Whisper** transcribes (encoder–decoder), **CLAP** retrieves (two towers, like CLIP). Same spectrogram tokens, different heads.

1. Whisper: listen, then write

Radford et al. (2022). The encoder is a transformer on log-mel spectrogram patches (note **6.1**). The decoder is causal and **cross-attends** to the encoder, like translation in note **3.3**. The output is text, plus special tokens for language, timestamps, and task (transcribe versus translate).

Whisper-style: audio encoder, text decoder (Week 3 block, new input)

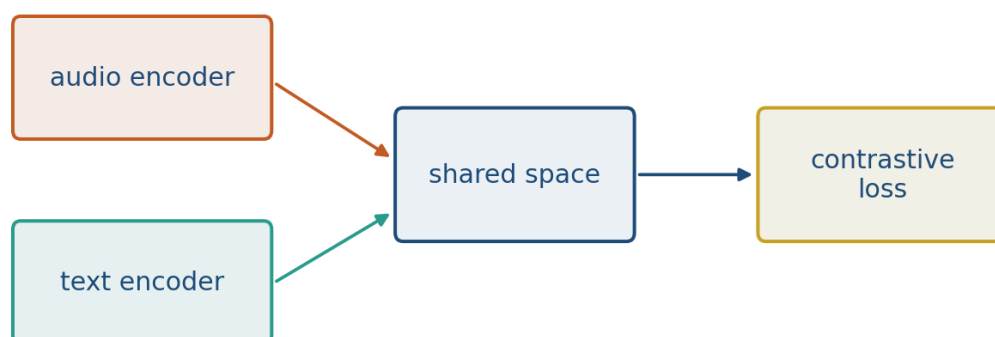


Pretraining is weakly supervised: a huge pile of audio paired with transcripts from the web, filtered rather than hand-labeled. Multilingual is the point. You do not train this from scratch in the lab; you would load a checkpoint. The architecture is what you need to draw.

2. CLAP: CLIP, but the image is sound

CLAP (Contrastive Language–Audio Pretraining) is coordinated, not joint. An audio encoder (CNN or transformer on the spectrogram) and a text encoder emit vectors in one space. InfoNCE on audio–caption pairs. After training you keep both towers.

CLAP is CLIP for audio: two towers, cosine, paired captions



Zero-shot tagging: encode "a dog barking", rank clips by cosine. Retrieval both ways. It is not a transcriber. If you need words, use Whisper (or a CTC ASR). If you need "find the clip that matches this sentence," use CLAP.

3. Which one in a project

Need	Model
Transcript, subtitles, timestamps	Whisper
Search / zero-shot tags / audio-text retrieval	CLAP
Generate a caption in free text	a decoder with an audio encoder (Whisper-like, or a BLIP-style audio LM)

Fusing audio with vision or text is next (note **6.3**). Do not concatenate a Whisper hidden state into CLIP and call it a paper; say how you aligned the rates and the losses.

4. Practice

1. Whisper's decoder is causal. Why can CLAP's text tower be bidirectional like BERT?
2. You have a gallery of podcast clips and a typed query. Whisper or CLAP? Why?
3. Name one special token Whisper uses that CLIP never needed.